

COMP6237 – Linear Regression

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Lecture slides available here:

<http://comp6237.ecs.soton.ac.uk/>

(Credit goes to Jon Hare who developed a large part of the module)

General Plan

- At the start of the course:
 - An introduction to regression techniques
- These lectures are based on the stats package R
 - This is free software, you can download it from <https://www.r-project.org/>
 - If you are not familiar with R, follow a tutorial to get some idea:
<https://www.southampton.ac.uk/~mb1a10/Rtutorial/R.html>
- At the end of the course:
 - An introduction to Information Theory
 - Mining Data Streams

COMP6237: Linear Regression

.Outline:

- Brief revision of some basic stats
- Variables and prediction
- The method of least squares (LS)
- Practical implementation in R
- Linear regression in higher dimensions
- Maximum likelihood estimation (MLE)
- LS and MLE
- Weighted LS, Heteroskedasticity, and local linear regression
- Problem sheet for linear regression and MLE:
– <https://www.southampton.ac.uk/~mb1a10/stats/exercise1.pdf>

Reminder of Some Basic Stats (1)

- .Suppose we have a set of N observations/measurements
- .Can analyse them via histograms/pdfs $\{X_1, X_2, \dots, X_N\}$
- .How to classify distributions?
 - Means (Median, mode, etc.) – measures of central tendency for a distribution

$$E[X] = 1/N \sum_{i=1}^N X_i$$

- Variances/standard deviation – measures of dispersion

$$V[X] = 1/N \sum_{i=1}^N (X_i - E[X])^2 = E[X^2] - E[X]^2$$

- Could use the Central Limit Theorem to argue about standard errors, confidence intervals, etc.

Reminder of Some Basic Stats (2)

• We are often interested in relationships between pairs of observations

$$\{(X_1, Y_1), (X_2, Y_2), \dots, (X_N, Y_N)\}$$

• Can classify relationships in various ways

- Covariance (measure of the joint variability of two random variables)

$$\text{Cov}[X, Y] = 1/N \sum_{i=1}^N (X_i - E[X]) (Y_i - E[Y])$$

- Correlation coefficients, e.g. Pearson

$$r = \frac{\text{Cov}[X, Y]}{\sqrt{V[X]} \sqrt{V[Y]}}$$

- R^2 ... proportion of variance of X explained by knowledge of Y

What about Prediction?

- What about if we want to predict the value of one variable based on the knowledge about another variable?
- This is what regression analysis is all about

Interlude: Types of Variables (1)

.Three types of variables:

- **Continuous**: real numbered values (e.g., time, mass)
- **Ordinal**: a numerical variable where a small number of possibilities are ranked (e.g., school grades)
 - . Boundaries between first two types can be blurred (e.g. most “continuous” variables are really ordinal because they can only be measured up to some accuracy)
- **Categorical**: describes membership of a group (but cannot be ranked). e.g., country of birth, gender, etc.
 - . Some are binary and others are not

Interlude: Types of Variables (2)

.Whatever form a variable has, they can play different roles when we build statistical models

.Dependent or outcome variables

- For our analysis this variable is the focus. It is assumed to be predictable from some other variables.

.Independent or predictor variables

- Are assumed to have inherent variation (“they just are”). We will use them to explain the variance in the dependent variables.

Example: Demographic Factors

.Let's assume we want to predict driving test outcomes from demographic data.

.For this particular analysis:

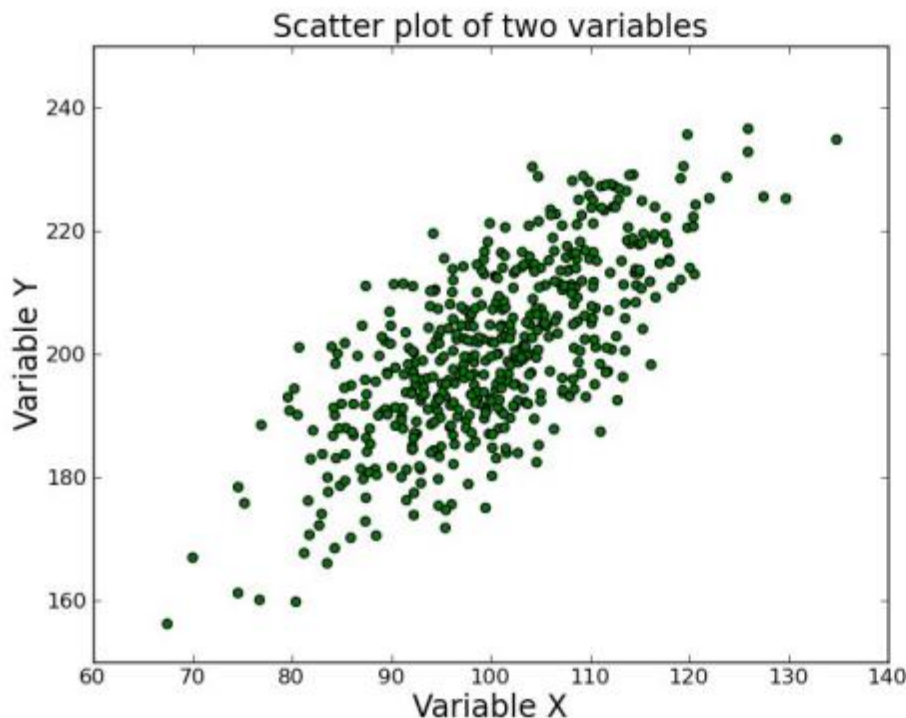
- Independent variables: demographic factors like gender, age, weight, height, etc.
- Dependent variables: driving test outcomes like outcome of the practical/theoretical parts

.Role of variables depends on analysis/model we have in mind.

- We could equally well be interested in the inverse, i.e., predicting demographic factors based on knowledge of driving test outcomes.

Variables in Regression

- Simplest case involves one dependent (Y) and one independent (X) variable
- Both variables are continuous (or at least ordinal)
- We are trying to predict the variation in Y based on the variation in X

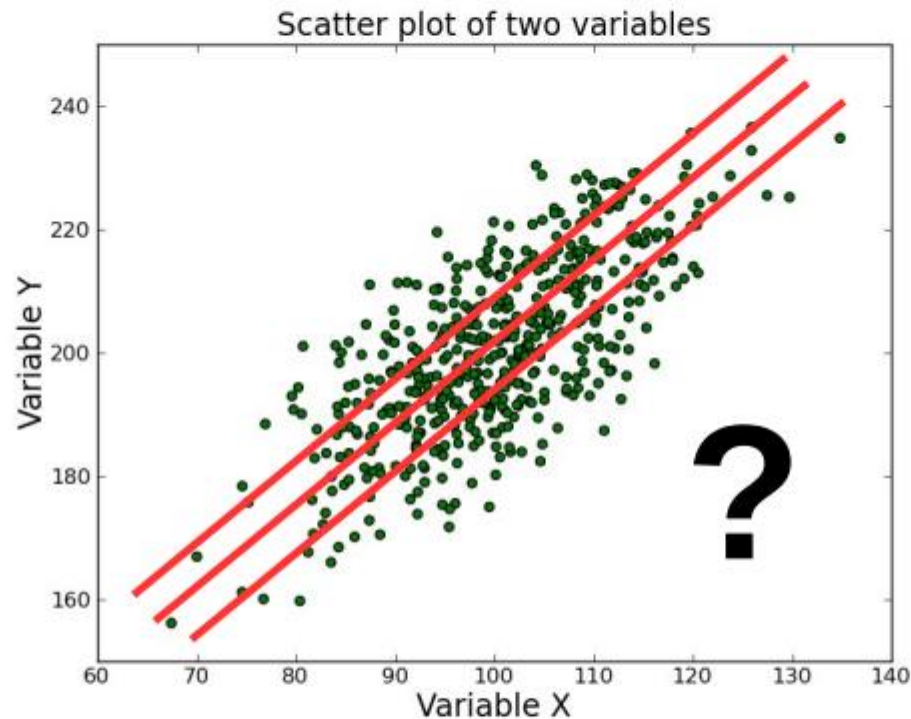


Given a plot like this:
What are your options?

Easiest way:
Assume linear relationship
between X and Y

Try to find line of “best guess”

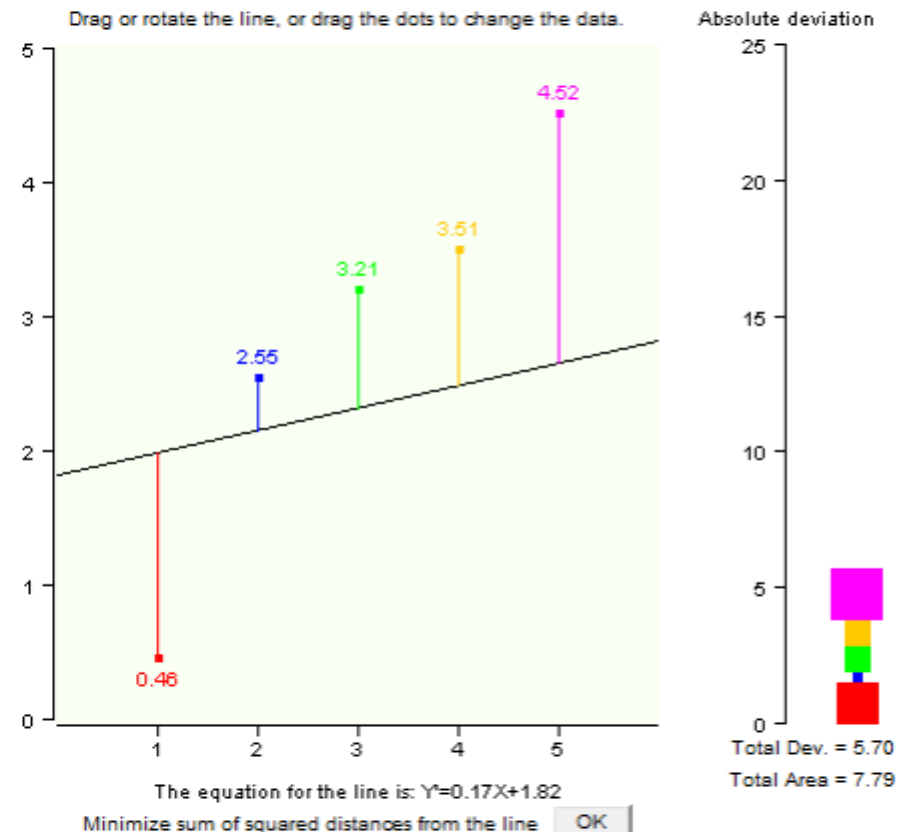
Drawing a Line Through Some Points



- Look for line of “best fit” through cloud of X,Y points
- $Y=mX+b$, Need to find two parameters m and b
- How to be systematic about it?

Method of Least Squares

- For any given line $mX+b$ we can measure the difference between our actual Y values and those predicted
- **Squared differences** are a reasonable measure of the goodness of fit; regression analysis tries to minimize this difference



LS as an Optimisation Problem

.One might think that minimizing the squared differences is a difficult combinatorial optimisation problem ... it is not:

$$E(m, b) = \sum_{i=1}^N (mX_i + b - Y_i)^2 = \min!$$

$$\frac{\partial}{\partial b} E(m, b) = 2 \sum_{i=1}^N (mX_i + b - Y_i) = 0$$

$$b = E[Y] - mE[X]$$

.Substituting back:

$$E(m, b) = \sum_{i=1}^N (m(X_i - E[X]) - (Y_i - E[Y]))^2$$

$$\frac{\partial}{\partial m} E(m, b) = 2 \sum_{i=1}^N (m(X_i - E[X]) - (Y_i - E[Y])) (X_i - E[X]) = 0$$

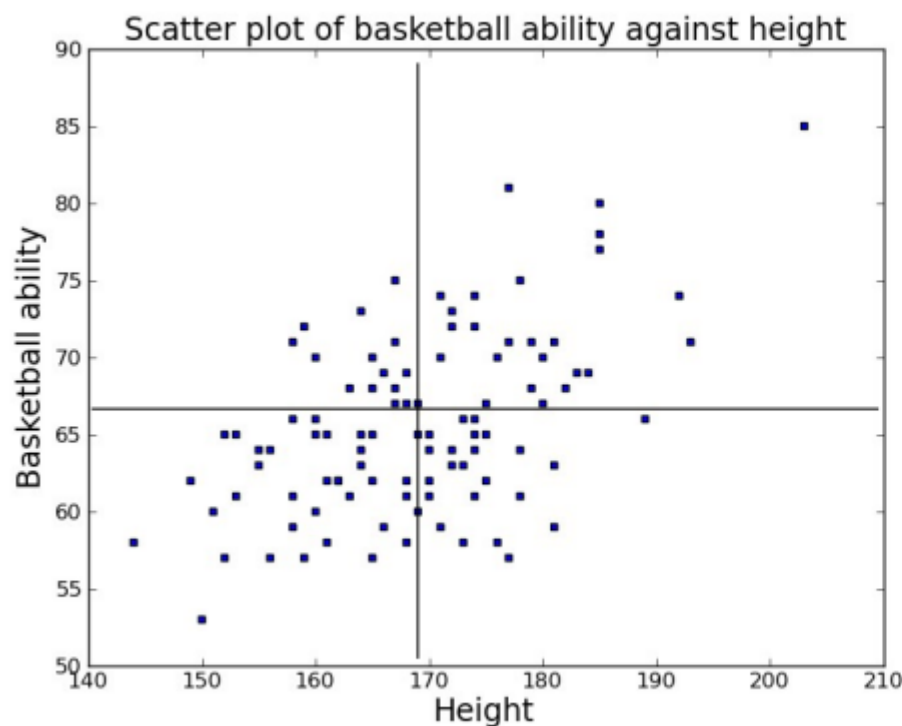
$$m = \frac{\sum_{i=1}^N (Y_i - E[Y]) (X_i - E[X])}{\sum_{i=1}^N (X_i - E[X])^2} = \text{Cov}[X, Y] / V[X]$$

Regression in Machine Learning

- Slightly different point of view is that we might consider the pairs (X_i, Y_i) as a training set
- Want to learn a mapping $y=f(x)$ from the training set
- Approach often:
 - “somehow” parametrise $f(x)$ (e.g., $f(x)=mx+b$ here) and try to learn best parameters m and b
 - This is often done via minimising some error function E which sums up squared residual errors over training set (i.e., this is the same as above)
 - e.g., in NN $f(x)$ is a nonlinear function

How Do I Run Linear Regression in R?

- Let's start with a fictional data set
 - Say a 100 men have been measured for their height and basketball ability
 - We want to predict basketball skill from height



Mean height is 169cm, stddev 10.5cm
Mean ability is 66 (arbitrary scale), stdev 6

Correlation $r=0.52$ (i.e. 28% of the variation of basketball ability are explained by height)

R Commands ...

- Read data into R with the usual command `read.table ()`; the variables of interest are Height and BasketballAbility
- Build a regression model with
- `regmodel=lm (BasketballAbility ~ Height)`
- (lm stands for “linear model”)
- `summary (regmodel)` gives us most of the information we need ...

The Full Output

Residuals:

Min	1Q	Median	3Q	Max
-11.0733	-3.4851	-0.5733	3.4969	12.9267

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	15.67832	8.22351	1.907	0.0595	.
Height	0.29602	0.04855	6.097	2.15e-08	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.07 on 98 degrees of freedom

Multiple R-squared: 0.275, Adjusted R-squared: 0.2676

F-statistic: 37.17 on 1 and 98 DF, p-value: 2.146e-08

The Full Output

Residuals:

Min	1Q	Median	3Q	Max
-11.0733	-3.4851	-0.5733	3.4969	12.9267

Characterizes the distribution of “residuals” (differences between predicted output and actual output)

Should roughly be normally distributed with a mean of zero

The Full Output

This is the important test here, i.e. the hypothesis of a relationship (slope $\neq 0$) is confirmed.

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The coefficients (“m” and “b”).

Meaning ... a man of zero height would have basketball ability 15.7,
Every additional cm of height adds 0.3 to basketball ability.

Also get standard errors, t-tests for hypothesis that values are 0

Generalisations

- Multi-dimensional input?

- “fit” a hyperplane

$$\mathbf{x} \in \mathbb{R}^D, y \in \mathbb{R}$$

- A general linear regression function then is

$$f(\vec{x}) = \sum_{j=1}^D w_j x_j + b = \mathbf{w}\mathbf{x} + b$$

- For simpler notation we could say

$$\tilde{\mathbf{w}} = [w_1, w_2, \dots, w_D, b]^T, \tilde{\mathbf{x}} = [x_1, x_2, \dots, x_D, 1]^T$$

- Hence:

$$E = \sum_{i=1}^N (y_i - \tilde{\mathbf{w}}\tilde{\mathbf{x}}_i)^2$$

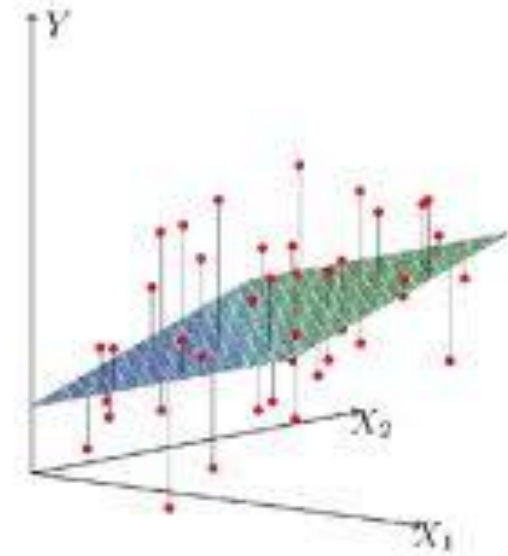


Figure 3.1: Linear least squares fitting with $X \in \mathbb{R}^2$. We seek the linear function of X that minimizes the sum of squared residuals from Y .

Generalisations (2)

- For a more compact notation say

$$\tilde{X} = \begin{bmatrix} \tilde{\mathbf{x}}_1^T & . \\ \tilde{\mathbf{x}}_2^T & . \\ . & . \\ . & . \\ \tilde{\mathbf{x}}_N^T & . \end{bmatrix} \quad \mathbf{y} = [y_1, y_2, \dots, y_N]^T$$

- Then: $E = \sum_{i=1}^N (y_i - \tilde{\mathbf{w}} \tilde{\mathbf{x}}_i)^2 = \|\mathbf{y} - \tilde{X} \tilde{\mathbf{w}}\|^2 = (\mathbf{y} - \tilde{X} \tilde{\mathbf{w}})^T (\mathbf{y} - \tilde{X} \tilde{\mathbf{w}})$
- Can find $\mathbf{w}$ through $dE/d\mathbf{w}_i = 0 \dots$

$$\tilde{\mathbf{w}} = (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T \mathbf{y} = X^+ \mathbf{y}$$



pseudoinverse of X also denoted by X^+

Generalisations (3)

- Some practical considerations

- The data matrix X can have large dimensions and we might need to calculate an inverse ... this can be quite time consuming

- Two ways to go about it:

- Often via **QR factorization**, factorizing $\tilde{X} = QR$
- with Q a diagonal matrix and R an upper triangular matrix (can be done via applying Gram-Schmidt procedure to column vectors) → for exact solutions
- Alternatively: can use **stochastic gradient descent** to numerically minimize residuals

Generalisations (4)

- D dimensional input, K dimensional output; could say:

$$\mathbf{y} = \tilde{\mathbf{W}}^T \tilde{\mathbf{x}} \quad \tilde{\mathbf{W}} = [\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \dots, \tilde{\mathbf{w}}_K] = \begin{bmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \dots & \mathbf{w}_K \\ b_1 & b_2 & \dots & b_K \end{bmatrix}$$

This is useful since: $y_j = \tilde{\mathbf{w}}_j^T \tilde{\mathbf{x}}$

- An error function can be constructed by summing over all pairs and output dimensions

$$E = \sum_{i=1}^N \sum_{j=1}^K (y_{i,j} - \tilde{\mathbf{w}}_j^T \tilde{\mathbf{x}}_i)^2$$

- Introduce: $\mathbf{y}'_j = [y_{1,j}, y_{2,j}, \dots, y_{N,j}]^T$ $\tilde{\mathbf{X}}^T = [\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2, \dots, \tilde{\mathbf{x}}_N]$

$$\rightarrow E = \sum_{j=1}^K \|\mathbf{y}'_j - \tilde{\mathbf{X}} \tilde{\mathbf{w}}_j\|^2$$

- i.e. we have K distinct estimation problems with solutions $\tilde{\mathbf{w}}_j = \tilde{\mathbf{X}}^+ \mathbf{y}'_j$

Some Comments

.One may have wondered if different results would have been obtained when making a different choice about the difference function between predictions and data points?

- i.e., what if one wouldn't use the sum of the squares ($\sim L_2$ norm) but some other measure?
→ other results would have been obtained!

.Least squares is a very common strategy in the sciences

.Another prominent approach is maximum likelihood estimates

Maximum Likelihood Estimation

- Suppose we have a set of n data points $X_1, \dots, X_n$ which are from some pdf $f(x;p)$ which has some “hidden” parameters p . We want to “guess” these parameters.
- How to go about it? Construct a likelihood function:

$$L(X_1, X_2, \dots, X_n; p) = \prod_{i=1}^n f(X_i; p)$$

Likelihood of obtaining the data by sampling from f given the parameter p

- Maximising L will give us a value of p corresponding to the most likely explanation of the data

Example (1)

• Suppose our observations have been generated from a normal distribution with unknown mean and variance, i.e.,

$$X \sim N(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(\frac{-(x - \mu)^2}{2\sigma^2}\right)$$

• Then the likelihood function is:

$$L(X_1, \dots, X_n; \mu, \sigma^2) = \sigma^{-n} (2\pi)^{-n/2} \exp\left(\frac{-1}{2\sigma^2} ((X_1 - \mu)^2 + (X_2 - \mu)^2 + \dots + (X_n - \mu)^2)\right)$$

• L is max if $\sum_{i=1}^N (X_i - \mu)^2 = \min!$ which is if $\mu = 1/N \sum_{i=1}^N X_i$

• Hence the expectation is a **maximum likelihood estimator** for this example.

Example (2)

.Suppose we have n observations $X_1, \dots, X_n$ that have been sampled from the uniform distribution over $[0, N]$ and N is unknown.

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.Construct likelihood function

$$L(X_1, \dots, X_n; N) = \begin{cases} 0 & \text{any } X_i \text{ outside } [0, N] \\ (1/N)^n & \text{otherwise} \end{cases}$$

.Maximum likelihood estimator?

Example (2)

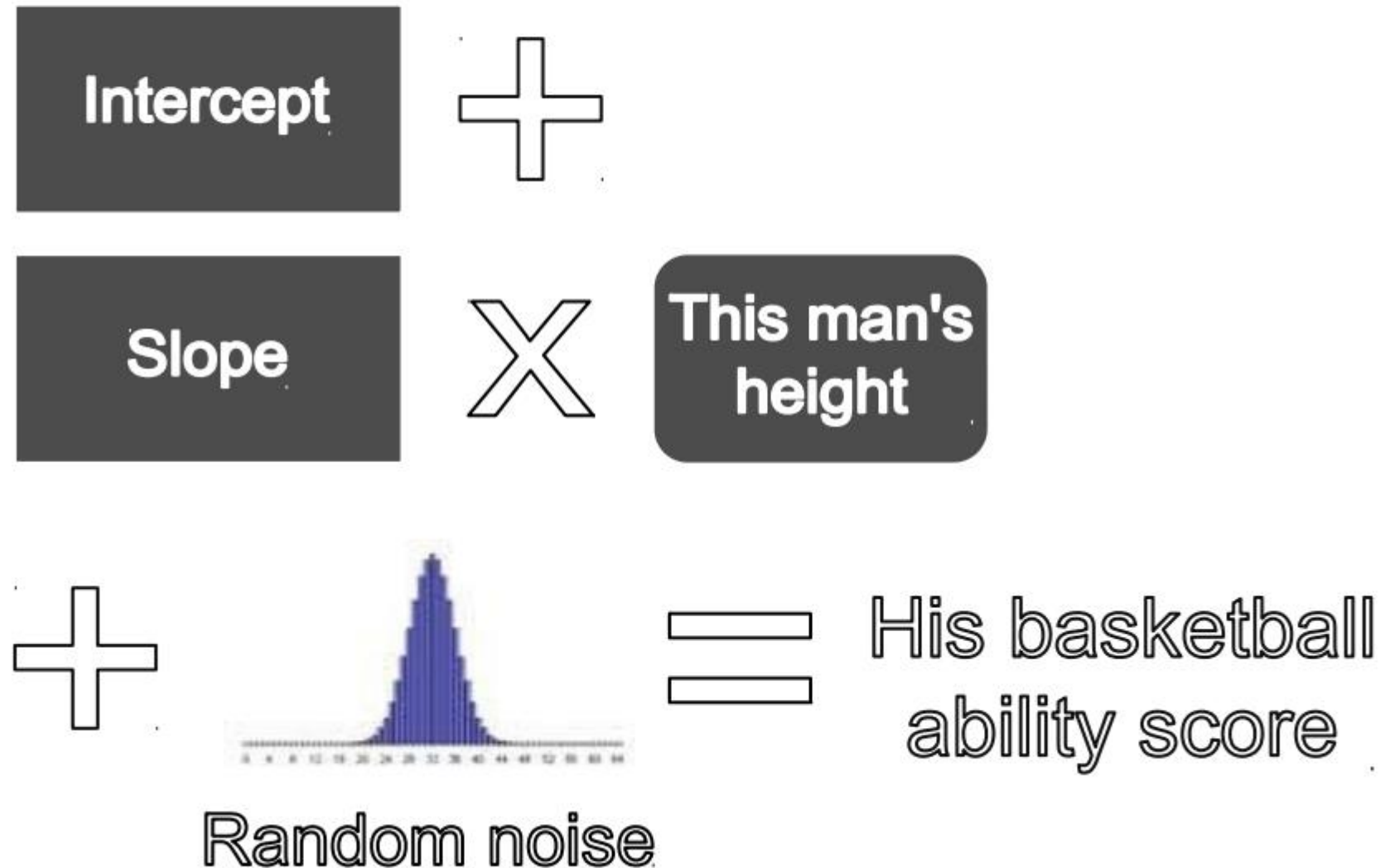
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.Maximum likelihood estimator is $N = \max(X_1, X_2, \dots, X_n)$

Implied Model of Data Generation in Linear Regression



MLE and LS

• Suppose we know a priori that X and Y have a linear relationship except for some noise, i.e.,

$$Y_i = mX_i + b + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$$

• where the epsilons are independent. We aim to find m and b via MLE:

$$L(X_1, \dots, X_n; m, b) = (2\pi)^{-n/2} \sigma^{-n} \exp\left(\frac{-1}{2\sigma^2} \sum_i (Y_i - mX_i - b)^2\right)$$

• To maximise L we need to minimise the sum of squares in the exponent, i.e.,

- Least squares is equivalent to an MLE estimate for m and b if X and Y are **linearly** related with **Gaussian noise**. Sum of squares has a privileged position ...

Weighted Least Squares

• Instead of minimising

$$\sum_{i=1}^N (mX_i + b - Y_i)^2 = \min!$$

• one might want to minimise:

$$\sum_{i=1}^N w_i (mX_i + b - Y_i)^2 = \min!$$

• Why?

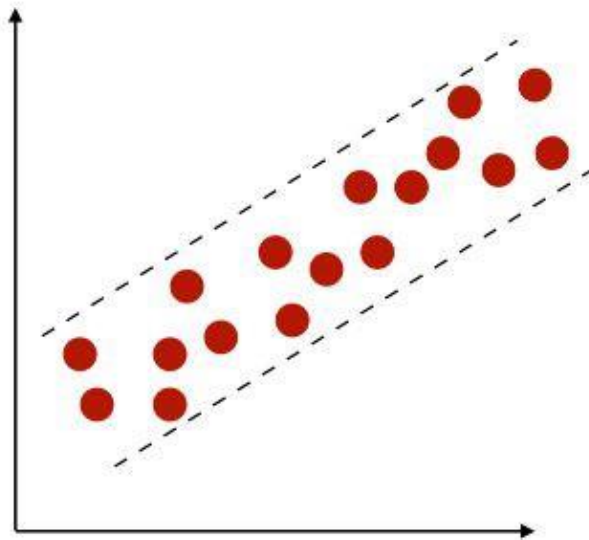
- To focus accuracy – might be more interested in certain X-regions, or errors in some regions might be more costly than in others
- There is a number of other optimisation problems transformed/approximated by WLS, e.g., generalised linear models where the response is some nonlinear function of a linear predictor (e.g., logistic regression, see later)

Homo-/Heteroskedasticity

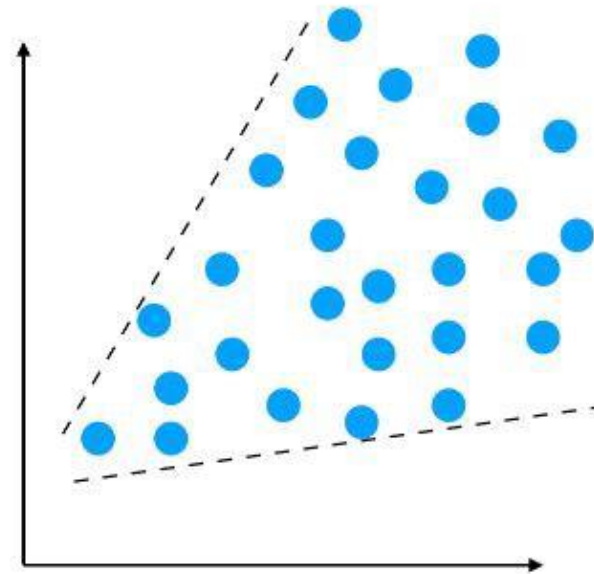
.Discounting imprecision

- Ordinary least squares assumes $y=mx+b+e$ with e iid Gaussian white noise. This implies that e has constant variance (**homoskedasticity**).
- Often this is not the case → **heteroskedastic data**
- Can then set $w_i=1/s_i^2$ so we get the heteroskedastic MLE
- (in other words: does not make much sense to concentrate on noisy parts of the data, want to use parts with little noise for our estimates)

Homo-/Heteroskedasticity – Examples



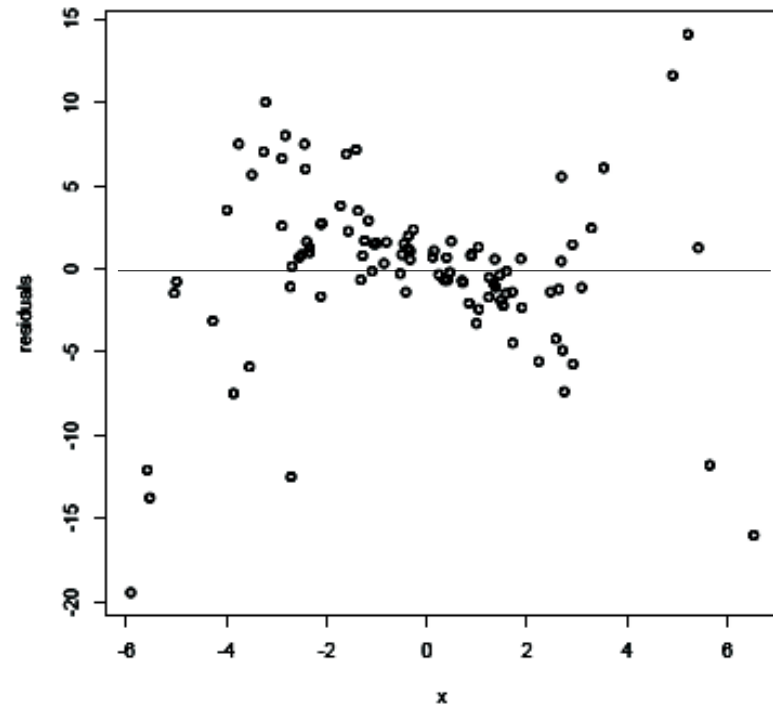
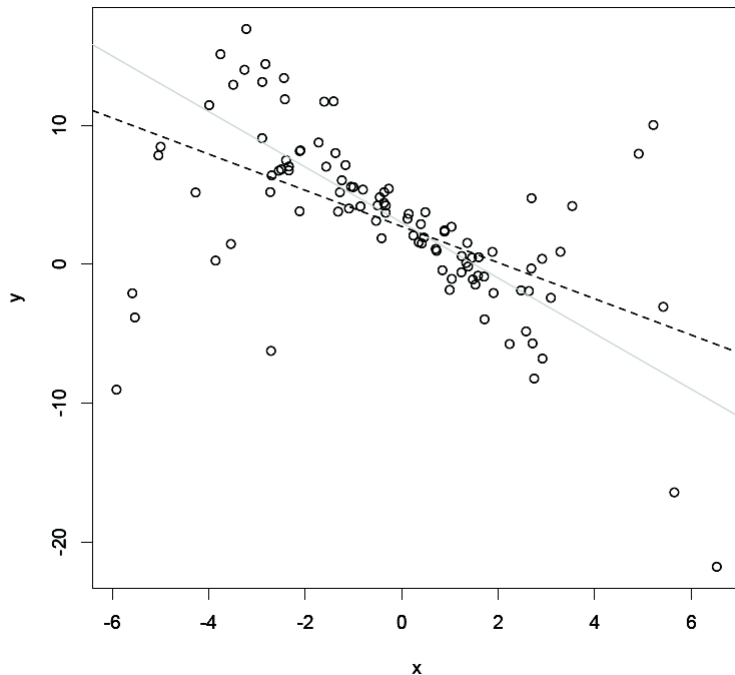
Homoscedasticity



Heteroscedasticity

An Example in R

.Suppose we have a linear relationship $y=3-2x$ between X and Y , and add Gaussian white noise with $s(x)=1+0.5x^2$

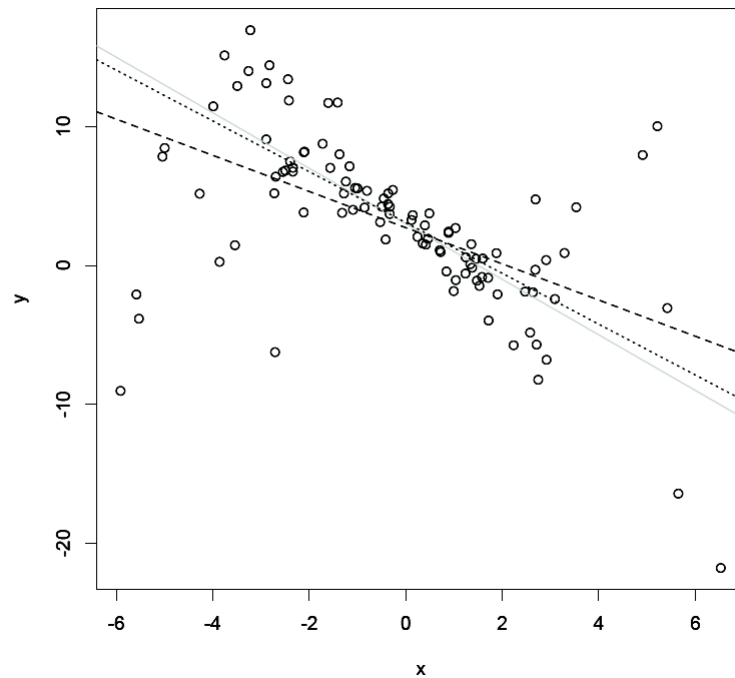


```
x=rnorm(100,0,3)
Y=3-2*x+rnorm(100, 0, sqrt(1+0.5*x^2))
plot(x,y)
abline(a=3,b=-2,col="grey")
fit.ols= lm(y~x)
abline(fit.ols$coefficients,lty=2)
```

plot(x,fit.ols\$residuals)

As expected, variance is not constant!
Fit misses the real relationship by some margin.

Weighted Linear Regression



unweighted regression

weighted regression performs better

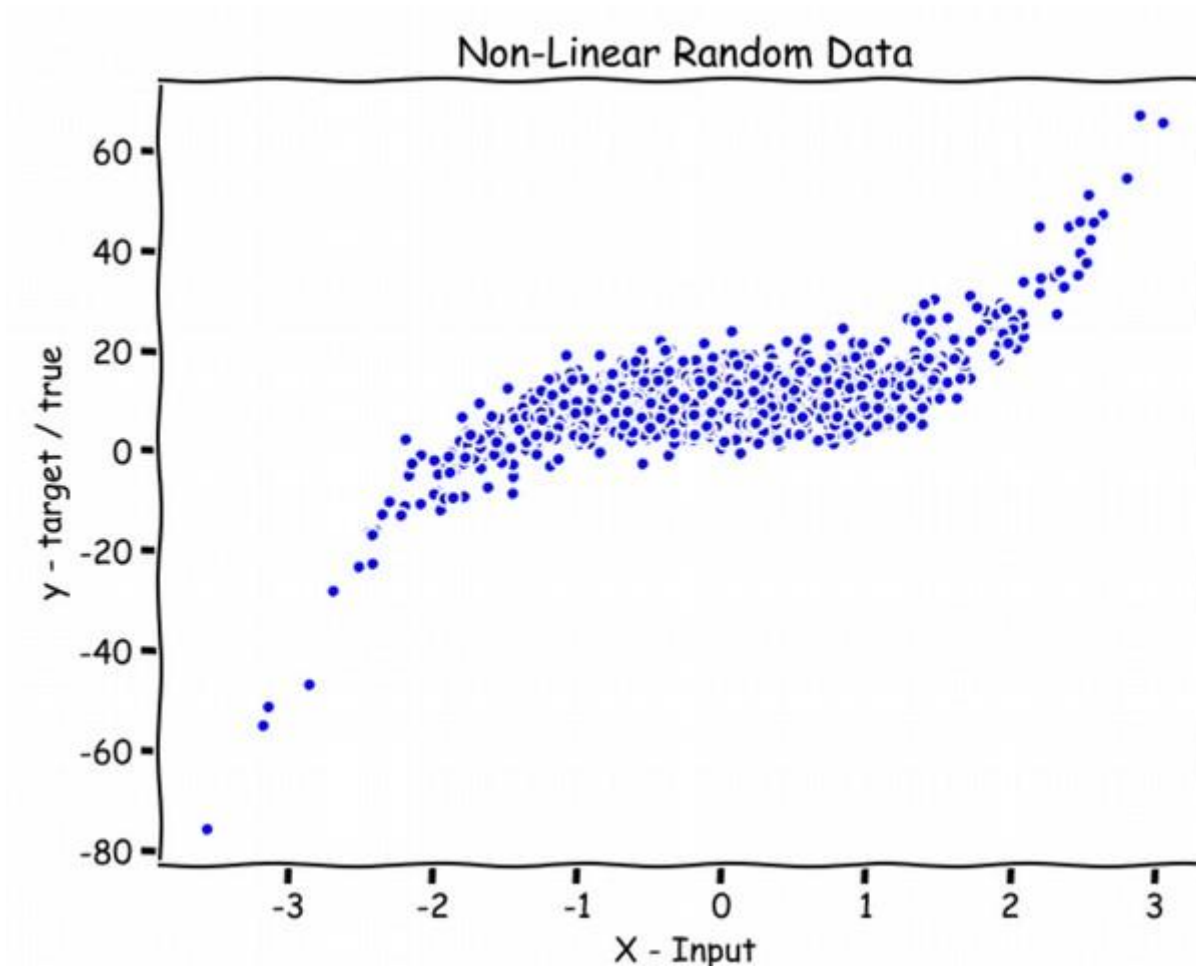
real relationship

```
fit.wls = lm (y~x, weights=1/(1+0.5*x*x))  
abline(fit.wls$coefficients, lty=3)
```

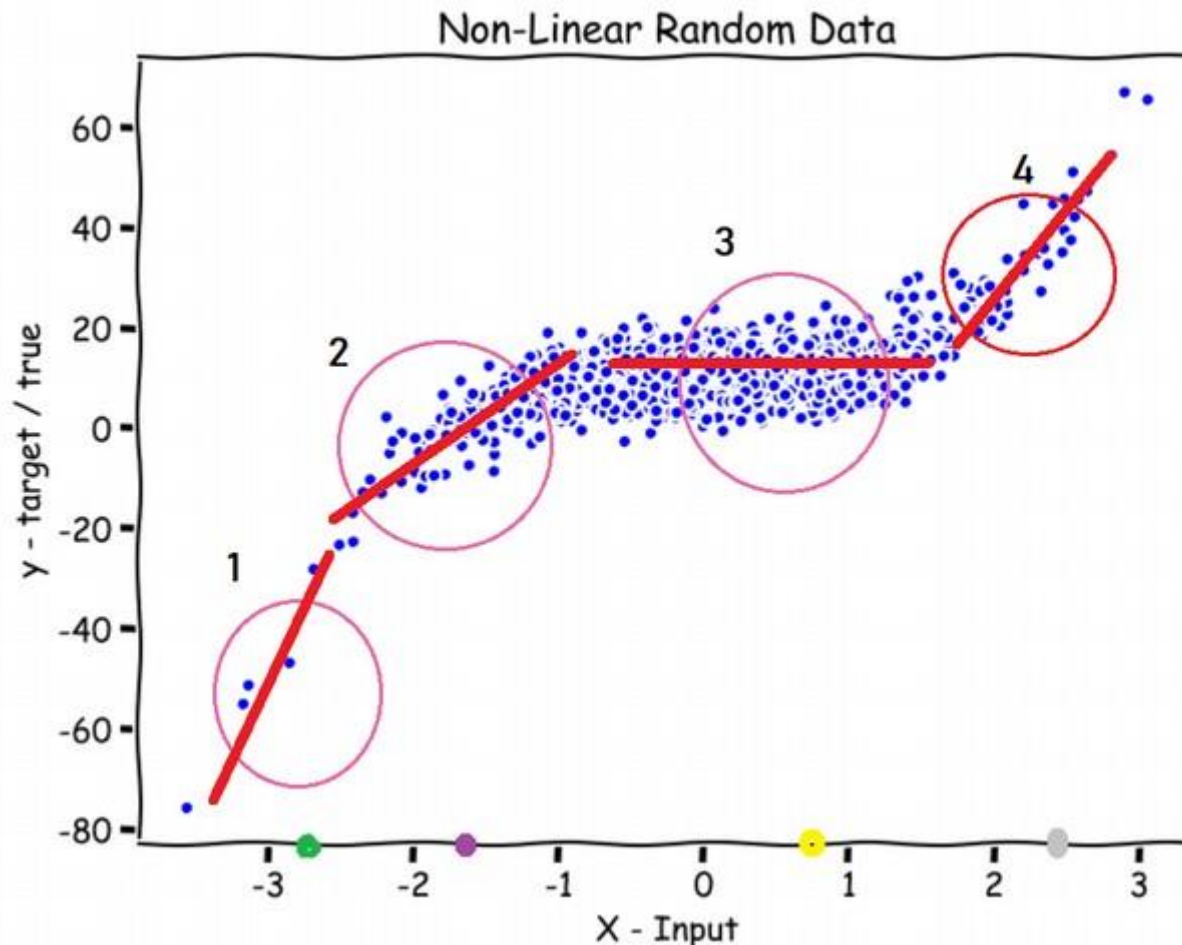
How to know the proper weights?

- Somehow know it from measurement device (e.g., know precision of the device for various ranges)
- e.g., in polls or surveys variance of the proportions we find should be inversely related to sample size, hence can make weights proportional to sample size.
- Try to estimate it from the data, e.g.
 - Estimate $y(x)$
 - Construct log squared residuals $z_i = \log((y_i - r(x_i))^2)$
 - Estimate mean of the z 's $\rightarrow q(x)$
 - Use $s_x^2 = \exp q(x)$

What if the data are not linear?



Local linear regression



Local Linear Regression

- Linear regression could be justified by looking at a general regression function $r(x)$ and expanding into a Taylor series

$$r(x) = r(x_0) + (x - x_0)r' + 1/2(x - x_0)^2 r''(x_0) + \dots (*)$$

- Then as long as we are close to x_0 r' is the regression coefficient, but what if the relationship is not linear?
- Naive approach: use some window (say of size h) around data points and set

$$w_i = \begin{cases} 1 & \text{if } |x_i - x_0| < h \\ 0 & \text{otherwise} \end{cases}$$

... and then use weighted least squares

Local Linear Regression (2)

.Often one wants weights to change a bit more smoothly than that.

.Kernel regression:

– Cut off Taylor expansion (*) after constant term and solve

$$\min_b \sum_{i=1}^N w_i(x) (y_i - b)^2$$

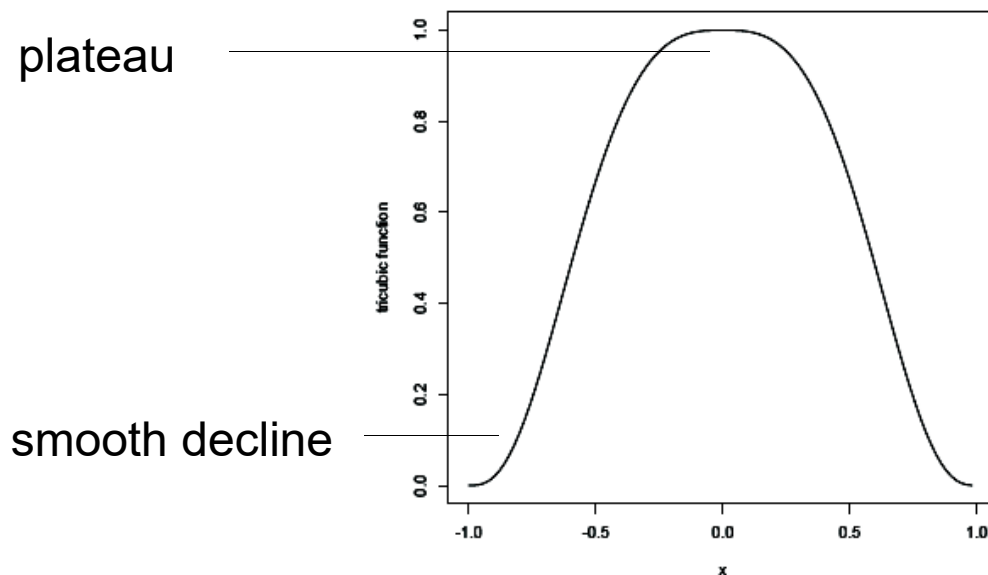
– Which is solved by

$$b = \frac{\sum_{i=1}^N w_i(x) y_i}{\sum_{i=1}^N w_i(x)}$$

$$w_i(x) \propto K(x_i, x)$$

Locally Linear Regression (3)

- Take 0st and 1st order terms from (*)
- Often use a tri-cubic kernel

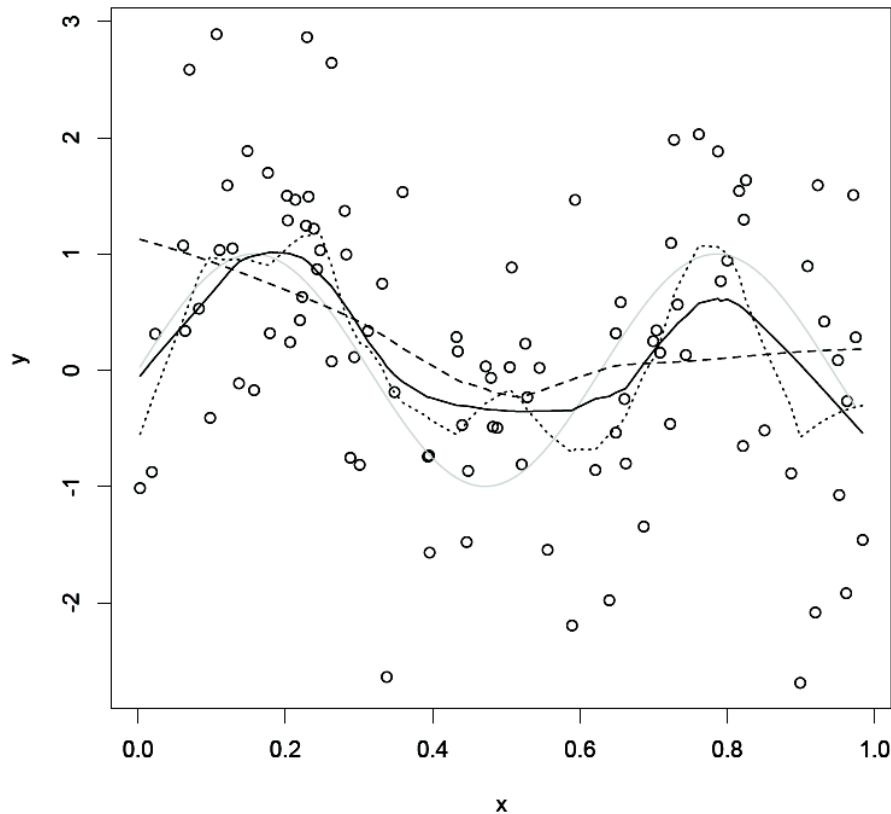


$$K(x_i, x) = \left(1 - \left(\frac{|x - x_0|}{h}\right)^3\right)^3$$

(h=1)

- R functions: lowess (specify fraction f of data points included), loess

Example



```
x.sin=runif(100,0,1)
y.sin=sin(10*x.sin)+rnorm(100,0,1)
plot(x.sin,y.sin, xlab="x", ylab="y")
curve(sin(10*x),col="grey",add="TRUE")
lines(lowess(x.sin, y.sin,f=1/3),lty=1)
lines(lowess(x.sin, y.sin,f=2/3),lty=2)
lines(lowess(x.sin, y.sin,f=1/6),lty=3)
```

Some art in choosing f appropriately; but can do quite a good job.

Local Linear Regression (4)

.When would you actually use this?

- Number of predictors is small
- You don't want to think too hard about what features to use
- (we will talk about other ways to deal with non-linear data later)

. Cons:

- Linear regression is a parametric technique: estimate weights from data and can then use to predict
- This is a non-parametric technique (the “data provide the function” – basically need to keep data points in memory to make predictions)
- Need more data ...

Summary

.Linear regression

- What it is, how it works
- How to judge significance
- Generalisations to higher d's

.MLE

- Formalisation
- Relationship to LR

.Extensions of linear regression

- Homo vs heteroskedastic data
- Weighted linear regression

Problem (1)

.A data set is constructed by taking 100 samples from a normal distribution with mean 5 and standard deviation 2 to construct a variable X_i , $i=1, \dots, 100$. Then, a variable Y_i , $i=1, \dots, 100$ is constructed by taking the values of the corresponding X_i and adding one half of a random variate drawn from a normal distribution with mean 5 and standard deviation 2 and thus a set of 100 pairs (X_i, Y_i) is obtained.

.Q: Find the parameters of a linear regression of Y on X (both by doing the numerical experiment and by calculating the result analytically).

Problem (2)

.Some person wants to conduct a least squares regression on a data set of N (X,Y) pairs, but wants attaches varying importance to deviations of various (X,Y) pairs to the line of best fit. The relative importance of deviations of pair (X_i, Y_i) are given by a function $f(i)$. Find an expression for the line of best fit generated by this procedure.

Problem (3)

- Repeated coin tossing of an (unfair) coin produces 100 heads up and 120 tails up. Find a maximum likelihood estimate for the probability that a coin toss will result in heads up.
- N variables have been sampled from an exponential distribution with unknown parameter. Find an expression for a maximum likelihood estimate for the parameter characterising the exponential distribution.